

- Let $A = (a_{ij})$ be an $\ell \times m$ matrix and let $B = (b_{ij})$ be an $m \times n$ matrix. We define the $\ell \times n$ product matrix $AB = (c_{ij})$ with ij entry

$$c_{ij} = \sum_{k=1}^m a_{ik}b_{kj}.$$

Special case: If $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n = \mathbb{C}^{n \times 1}$ then $\mathbf{x}^* \mathbf{y}$ is the standard Hermitian inner product.

- Matrix multiplication can be partitioned in various ways. For example:

$$(ij \text{ entry of } AB) = (i\text{-th row of } A)(j\text{-th col of } B)$$

$$(i\text{-th row of } AB) = (i\text{-th row of } A)B$$

$$(j\text{-th col of } AB) = A(j\text{-th col of } B)$$

- A function $f : V \rightarrow W$ between vector spaces is called linear if $f(\sum_i \beta_i \mathbf{v}_i) = \sum_i \beta_i f(\mathbf{v}_i)$ for all vectors $\mathbf{v}_i \in V$ and scalars β_i . If $g : U \rightarrow V$ is also linear then the composite function $(f \circ g) : U \rightarrow W$ is linear.
- Let $\mathbf{e}_1, \dots, \mathbf{e}_m \in \mathbb{C}^m$ be the standard basis. For any linear function $f : \mathbb{C}^m \rightarrow \mathbb{C}^n$ we define the $n \times m$ matrix $[f] \in \mathbb{C}^{n \times m}$ whose j -th column is $f(\mathbf{e}_j)$. For any $\mathbf{x} \in \mathbb{C}^m$ we have $f(\mathbf{x}) = [f]\mathbf{x}$, and for any other linear function $g : \mathbb{C}^\ell \rightarrow \mathbb{C}^m$ we have $[f \circ g] = [f][g]$.
- The identity $[f \circ g] = [f][g]$ implies that matrix multiplication is associative.
- The identity matrix I_n has column vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$. For any $A \in \mathbb{C}^{m \times n}$ we have $AI_n = A$ and $I_m A = A$. It follows that $A\mathbf{e}_j$ is the j -th column of A , $\mathbf{e}_i^* A$ is the i -th row of A and $\mathbf{e}_i^* A\mathbf{e}_j$ is the ij entry of A .
- For any linear function $f : \mathbb{C}^m \rightarrow \mathbb{C}^n$ there exists a unique linear function $f^* : \mathbb{C}^n \rightarrow \mathbb{C}^m$ with the property that $\langle f(\mathbf{x}), \mathbf{y} \rangle = \langle \mathbf{x}, f^*(\mathbf{y}) \rangle$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$. In terms of matrices, $[f^*] = [f]^*$ is the conjugate transpose. We have $(AB)^* = B^* A^*$.
- For any matrix $U \in \mathbb{C}^{m \times n}$ we have $U^* U = I_n$ if and only if the columns of U are orthonormal, if and only if $\|U\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x} \in \mathbb{C}^n$, if and only if $\|U\mathbf{x} - U\mathbf{y}\| = \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$, if and only if $\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$. If $m = n$, such a matrix is called unitary. Real unitary matrices are called orthogonal.
- The rotation matrix $R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$ is orthogonal.
- For any vector nonzero vector $\mathbf{v} \in \mathbb{C}^n$, the matrix $P = \frac{1}{\|\mathbf{v}\|^2} \mathbf{v} \mathbf{v}^*$ corresponds to the orthogonal projection onto the line spanned by $\mathbf{v}$.